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Gain-adjustable amplifier circuit

Abstract

An amplifier circuit includes an amplifier for generating an amplified input signal according to an input signal, and an attenuator circuit coupled to the amplifier. The attenuator circuit includes an input terminal for receiving the input signal or the amplified input signal, an output terminal, a reference voltage terminal, a zeroth resistor-switch circuit, a first resistor-switch circuit, and a second resistor-switch circuit. The zeroth resistor-switch circuit includes a first terminal coupled to the input terminal, a second terminal coupled to the output terminal, a zeroth switch coupled to the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit, a zeroth resistor coupled between the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit, a first resistor coupled between the zeroth resistor and the second terminal of the zeroth resistor-switch circuit, and a first switch.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/294,857, filed on Dec. 30, 2021. Further, this application is a continuation-in-part of U.S. application Ser. No. 17/499,831, filed on Oct. 12, 2021. The contents of these applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention is related to an amplifier circuit, and more particularly to a gain-adjustable amplifier circuit.

BACKGROUND

(2) With increasing popularity of the Internet and mobile devices, users demand more for communication quality. However, the gain range of the traditional single-stage amplifier is limited. The traditional structure cannot meet the current practical application and design requirements. Therefore, it is an important issue to provide a gain-adjustable amplifier circuit.

SUMMARY

(3) An embodiment provides an amplifier circuit including an amplifier for generating an amplified input signal according to an input signal, and an attenuator circuit coupled to the amplifier. The attenuator circuit includes an input terminal for receiving the input signal or the amplified input signal, an output terminal, a reference voltage terminal, a zeroth resistor-switch circuit, a first resistor-switch circuit, and a second resistor-switch circuit. The zeroth resistor-switch circuit includes a first terminal coupled to one of the input terminal and the output terminal, a second terminal coupled to another one of the input terminal and the output terminal, a zeroth switch coupled to the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit, a zeroth resistor coupled between the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit, a first resistor coupled between the zeroth resistor and the second terminal of the zeroth resistor-switch circuit, and a first switch. The first switch includes a first terminal coupled to the zeroth resistor and the first resistor, a second terminal coupled to the second terminal of the zeroth resistor-switch circuit, and a control terminal. The first resistor-switch circuit includes a first terminal coupled to one of the input terminal and the reference voltage terminal, a second terminal coupled to another one of the input terminal and the reference voltage terminal, a zeroth resistor coupled between the first terminal of the first resistor-switch circuit and the second terminal of the first resistor-switch circuit, a first resistor coupled between the zeroth resistor and the second terminal of the first resistor-switch circuit, and a first switch. The first switch includes a first terminal coupled to the zeroth resistor and the first resistor, a second terminal coupled to the second terminal of the first resistor-switch circuit, and a control terminal. The second resistor-switch circuit includes a first terminal coupled to one of the output terminal and the reference voltage terminal, a second terminal coupled to another one of the output terminal and the reference voltage terminal, a zeroth resistor coupled between the first terminal of the second resistor-switch circuit and the second terminal of the second resistor-switch circuit, a first resistor coupled between the zeroth resistor and the second terminal of the second resistor-switch circuit, and a first switch. The first switch includes a first terminal coupled to the zeroth resistor and the first resistor, a second terminal coupled to the second terminal of the second resistor-switch circuit, and a control terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a gain-adjustable amplifier circuit of a first embodiment of the

present invention.

(2) FIG. 2 is a block diagram of a gain-adjustable amplifier circuit of a second embodiment of the present invention.

(3) FIG. 3 is a block diagram of a gain-adjustable amplifier circuit of a third embodiment of the present invention.

(4) FIG. 4 is a circuit diagram of the gain-adjustable amplifier circuit **1** of FIG. 1.

(5) FIG. 5 is a circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit of FIG. 1, 2 or 3.

(6) FIG. 6 is another circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit of FIG. 1, 2 or 3.

(7) FIG. 7 is another circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit of FIG. 1, 2 or 3.

(8) FIG. 8 is another circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit of FIG. 1, 2 or 3.

(9) FIG. 9 is another circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit of FIG. 1, 2 or 3.

DETAILED DESCRIPTION

(10) Below, exemplary embodiments will be described in detail with reference to accompanying drawings so as to be easily realized by a person having ordinary knowledge in the art. The inventive concept may be embodied in various forms without being limited to the exemplary embodiments set forth herein. Descriptions of well-known parts are omitted for clarity, and like reference numerals refer to like elements throughout.

(11) Please refer to FIG. 1. FIG. 1 is a block diagram of a gain-adjustable amplifier circuit of an embodiment of the present invention. The gain-adjustable amplifier circuit **1** includes an input terminal **11**, an output terminal **12**, amplifier **20** and an attenuator circuit **40**.

(12) The input terminal **11** is for receiving an input signal S_{in} , and the output terminal **12** is for sending an output signal S_{out} . The output signal S_{out} can be sent to the backend circuit for further process. The input signal S_{in} and the output signal S_{out} are, for example, but not limited to, radio frequency signals.

(13) An input terminal of the amplifier **20** can receive the input signal S_{in} through the input terminal **11** and can amplify the input signal S_{in} according to its gain, so as to output the amplified input signal $S_{in'}$ at an output terminal of the amplifier **20**. The amplifier **20** includes a low-noise amplifier or a power amplifier.

(14) An input terminal of the attenuator circuit **40** may be coupled to the output terminal of the amplifier **20**. The output terminal may be coupled to the output terminal **12**. The attenuator circuit **40** can provide a plurality of attenuations to the amplified input signal $S_{in'}$, and to generate the attenuated signal S_{at1} accordingly. For example, the attenuator circuit **40** can select one of a plurality of attenuations according to the gain required by the amplifier circuit **1** to attenuate the amplitude of the amplified input signal $S_{in'}$ to a specific value, thereby generating the attenuated signal S_{at1} . The output signal S_{out} can be generated by the attenuated signal S_{at1} . In this way, the gain of the amplifier circuit **1** can be related to the attenuation of the amplified input signal $S_{in'}$. That is, the gain required by the amplifier circuit **1** can be realized based on the attenuation of the amplified input signal $S_{in'}$. The gain of the amplifier circuit **1** may be 0 dB or greater than 0 dB. It should be noted that since the attenuator circuit **40** can provide a plurality of attenuations to the amplified input signal $S_{in'}$, the gain of the amplifier circuit **1** is not limited by the gain range of the amplifier **20**. Through the attenuator circuit **40** the gain range of the amplifier circuit **1** can be expended and the gain of the amplifier circuit **1** can be adjusted. Not only that, since the attenuator circuit **40** is coupled to the output of the amplifier **20**, the attenuator circuit **40** would not easily generate loading effect at the input of the amplifier **20**. Thus it is hard to affect the performance of the amplifier circuit **1**, so the linearity and noise figure of the amplifier circuit **1** are maintained.

(15) Please refer to FIG. 2. FIG. 2 is a block diagram of a gain-adjustable amplifier circuit of a second embodiment of the present invention. The major difference between the embodiment of FIG. 2 and the embodiment of FIG. 1 is that the gain-adjustable amplifier circuit 2 of FIG. 2 includes an amplifier 20 and an attenuator circuit 50.

(16) The input terminal of the attenuator circuit 50 may be coupled to the input terminal 11, and the output terminal may be coupled to the output terminal 12. The attenuator circuit 50 can be configured to provide a plurality of attenuations to the input signal S_{in} , and generate the attenuated signal S_{at2} accordingly. For example, the attenuator circuit 50 can select one of the attenuations according to the gain required by the amplifier circuit 2 to attenuate the amplitude of the input signal S_{in} to a specific value, thereby generating the attenuated signal S_{at2} .

(17) Furthermore, the amplifier circuit 2 can operate in either amplifying mode or bypass mode. When the intensity of the input signal S_{in} is not strong enough and needed to be amplified by the amplifier 20, the amplifier 20 can be enabled. The amplifier circuit 2 can operate in an amplifying mode with the first gain. Further, the input signal S_{in} can be inputted from the input terminal 11 to the amplifier 20. The amplifier 20 can amplify the input signal S_{in} according to its gain and output the amplified input signal $S_{in'}$ at its output terminal. The output signal S_{out} can be generated by the amplified input signal $S_{in'}$. Therefore, the first gain is related to the gain of the amplifier 20. When the intensity of the input signal S_{in} is high enough that it does not need to be amplified by the amplifier 20, the amplifier 20 can be disabled. The amplifier circuit 2 can operate in the bypass mode with the second gain (equal or less than 0 dB). It should be noted that the disabled amplifier 20 can reduce unnecessary current consumption. Further, the input signal S_{in} is input to the attenuator circuit 50 from the input terminal 11 and the attenuator circuit 50 can attenuate the input signal S_{in} according to the selected attenuation to generate the attenuated signal S_{at2} . The output signal S_{out} can be generated by the attenuated signal S_{at2} . In this way, the second gain can be related to the attenuation of the input signal S_{in} . That is, the second gain required by the amplifier circuit 2 can be achieved based on the attenuation of the input signal S_{in} . The second gain may be less than the first gain. The first gain may be greater than 0 dB, and the second gain may be equal or less than 0 dB. It should be noted that since the attenuator circuit 50 can provide a plurality of attenuations to the input signal S_{in} , the gain range of the amplifier circuit 2 can be expanded and the gain of the amplifier circuit 2 can be adjusted through the attenuator circuit 50.

(18) Please refer to FIG. 3. FIG. 3 is a block diagram of a gain-adjustable amplifier circuit of a third embodiment of the present invention. The major difference between the embodiment of FIG. 3 and the embodiment of FIG. 2 is that the gain-adjustable amplifier circuit 3 of FIG. 3 may further include an attenuator circuit 40. The input terminal of the attenuator circuit 40 may be coupled to the output terminal of the amplifier 20, and the output terminal of the amplifier 20 may be coupled to the output terminal 12. The attenuator circuits 40 and 50 can respectively provide a plurality of attenuations to the amplified input signal $S_{in'}$ and respectively generate attenuated signals S_{at1} and S_{at2} accordingly.

(19) Comparing to the amplifier circuit 2 of the second embodiment, when the amplifier circuit 3 operates in amplifying mode, the attenuator circuit 40 can attenuate the amplified input signal $S_{in'}$ according to the selected attenuation to generate the attenuator signal S_{at1} . The output signal S_{out} can be generated by the attenuator signal S_{at1} . In this way, the first gain can be related to the attenuation of the amplified input signal $S_{in'}$. That is, in this embodiment, the first gain required by the amplifier circuit 3 can be realized based on the attenuation of the amplified input signal $S_{in'}$, and the second gain required by the amplifier circuit 3 can be realized based on the attenuation of the input signal S_{in} . Since the amplified input signal $S_{in'}$ is generated after the input signal S_{in} passing through the amplifier 20, the first gain may be greater than the second gain. The first gain may be equal or greater than 0 dB, and the second gain may be equal or less than 0 dB.

(20) Please refer to FIG. 4. FIG. 4 is a circuit diagram of the gain-adjustable amplifier circuit 1 of FIG. 1. The amplifier 201 and the attenuator circuit 401 of FIG. 4 may correspond respectively to

the amplifier **20** and the attenuator circuit **40** of FIG. **1**.

(21) As shown in FIG. **4**, amplifier **201** can include transistors **M1** and **M2**. The first terminal of the transistor **M1** is coupled to the output terminal of the amplifier **201**, and the control terminal is for receiving the operation signal **VGG**. The operation signal **VGG** is for enabling or disabling the amplifier **201**. The first terminal of the transistor **M2** is coupled to the second terminal of the transistor **M1**. The second terminal is coupled to the reference potential terminal **13**, and the control terminal is coupled to the input terminal of the amplifier **201** and can further receive the bias signal **Vbias**. The bias signal **Vbias** can bias the transistor **M2**. The reference voltage terminal **REF** may be, but is not limited to, a ground terminal in the system.

(22) The attenuator circuit **401** can include a plurality of attenuator units **ATU0-ATUn**, each attenuator unit **ATU0-ATUn** is coupled between the input terminal and the output terminal of the attenuator circuit **401**, and each attenuator unit **ATU0-ATUn** can respectively provide one of the plurality of attenuations. For example, the attenuator unit **ATU0** can provide attenuation of 0 dB; the attenuator unit **ATU1** can provide attenuation of 5 dB; the attenuator unit **ATUi** can provide attenuation of 10 dB and the attenuator unit **ATUn** can provide attenuation of 20 dB. In other words, attenuator units **ATU0-ATUn** can provide different attenuations. It should be noted that when one of the attenuator units (e.g., the attenuator unit **ATU0**) is enabled, the rest of the attenuator units (e.g., the attenuator units **ATU1-ATUn**) are disabled.

(23) One of the attenuator units (e.g., the attenuator unit **ATU0**) can include a switch **SW40**, and a control terminal of the switch **SW40** can receive the control signal **CS40**. The control signal **CS40** can turn on or off the switch **SW40**, thereby enabling or disabling the attenuator unit **ATU0** (e.g., turning on the switch **SW40** can enable the attenuator unit **ATU0**, thereby providing attenuation of 0 dB).

(24) The remaining attenuator units (e.g., the attenuator units **ATU1-ATUn**) of the plurality of attenuator units **ATU0-ATUn** may each include a π -type attenuator. Further, the π -type attenuator in the attenuator unit **ATU1** can include switches **SW411**, **SW421** and **SW431**. The control terminals of the switches **SW411**, **SW421** and **SW431** can receive the control signal **CS41**. The control signal **CS41** can turn on or off the switches **SW411**, **SW421** and **SW431**, thereby enabling or disabling the attenuator unit **ATU1**. For example, turning on the switches **SW411**, **SW421** and **SW431** can enable the attenuator unit **ATU1** to provide attenuation of 5 dB. Then π -type attenuator in the attenuator unit **ATUi** can include switches **SW41i**, **SW42i** and **SW43i**. The control terminals of the switches **SW41i**, **SW42i** and **SW43i** can receive the control signal **CS4i**. The control signal **CS4i** can turn on or off the switches **SW41i**, **SW42i** and **SW43i**, thereby enabling or disabling the attenuator unit **ATUi**. The π -type attenuator in the attenuator unit **ATUn** can include switches **SW41n**, **SW42n** and **SW43n**. The control terminals of switches **SW41n**, **SW42n** and **SW43n** can receive the control signal **CS4n**. The control signal **CS4n** can turn on or off the switches **SW41n**, **SW42n** and **SW43n**, thereby enabling or disabling the attenuator unit **ATUn**, and so on. It should be noted that “i” and “n” are positive integers. The control signals **CS40-CS4n** can be provided by an internal circuit of the attenuator circuit **401** or an external circuit outside of the attenuator circuit **401**. It should be noted that although the number of attenuator units shown in FIG. **4** is more than two, the actual number of attenuator units may be determined according to the actual application or design requirements. Hence, the invention is not limited to the number of attenuator units shown in FIG. **4**.

(25) In addition, the first terminal of the switch **SW40** is coupled to the input terminal of the attenuator circuit **401**, and the second terminal is coupled to the output terminal of the attenuator circuit **401**. The first terminal of each switch **SW411-SW41n** is coupled to the input terminal of the attenuator circuit **401**, and the second terminal is coupled to reference voltage terminal **REF**. The first terminal of each switch **SW421-SW42n** is coupled to the output terminal of the attenuator circuit **401**, and the second terminal of each switch **SW421-SW42n** is coupled to the reference voltage terminal **REF**. The first terminal of each switch **SW431-SW43n** is coupled to the input terminal of the attenuator circuit **401**, and the second terminal of each switch **SW431-SW43n** is

coupled to the output terminal of the attenuator circuit **401**. In other embodiments, the attenuator circuit **40** shown in FIG. 3 may also include the attenuator circuit **401** shown in FIG. 4.

(26) The attenuation corresponding to each attenuator unit ATU1-ATUn may be related to its resistance. For example, the on-resistance of the switches SW**411**, SW**421** and SW**431** can be chosen based on the attenuation corresponding to the attenuator unit ATU1. The on-resistance of the switches SW**41i**, SW**42i** and SW**43i** can be chosen based on the attenuation corresponding to the attenuator unit ATUi. The on-resistance of the switches SW**41n**, SW**42n** and SW**43n** can be chosen based on the attenuation corresponding to the attenuator unit ATUn, and so on. That is, the switches SW**411**-SW**43n** not only can switch the corresponding attenuator units ATU1-ATUn, but also can provide the corresponding attenuations. In other embodiments, the on-resistance of the switches SW**411** and SW**421** can be substantially the same; the on-resistance of the switches SW**41i** and SW**42i** can be substantially the same; the on-resistance of the switches SW**41n** and SW**42n** can be substantially the same, and so on. In another example, the attenuator units ATU1, ATUi and ATUn can provide 5 dB, 10 dB and 20 dB of attenuations respectively, and the on-resistance of the switch SW**431** can be less than the on-resistance of the switch SW**43i**; the on-resistance of the switch SW**43i** may be less than the on-resistance of the switch SW**43n**. In this embodiment, by properly choosing the channel width-to-length ratio of the switches SW**411**-SW**43n**, the switches SW**411**-SW**43n** can have appropriate on-resistance. In this way, when one of the attenuator units ATU1-ATUn is enabled, the appropriate corresponding attenuation can be provided.

(27) The amplifier circuit **1** may further include a matching circuit **60**, and capacitors AC**1**, AC**2** and AC**3**. The matching circuit **60** is coupled between the output terminal of the amplifier **201** and the input terminal of the attenuator circuit **401**, and is for providing an output impedance matching to the amplifier **201**. It can include resistors, inductors, capacitors or any combination of the foregoing three elements. Since the attenuator circuit **401** is substantially symmetrical, the difference between the impedance at the input terminal and the impedance at the output terminal of the attenuator circuit **401** can be within a predetermined range. Thus, the attenuator circuit **401** would not significantly impact the output impedance matching of the amplifier **201**. The capacitor AC**1** is coupled between the matching circuit **60** and the attenuator circuit **401**. The capacitor AC**2** is coupled between the attenuator circuit **401** and the output terminal **12**. The capacitor AC**3** is coupled between the attenuator circuit **401** and the reference voltage terminal REF. The capacitor AC**1** can block the DC bias voltage from the amplifier **201**, and the capacitors AC**2** and AC**3** can block the DC bias voltage from the attenuator circuit **401**. In other embodiments, the amplifier circuit **1** may selectively include capacitor AC**1**, AC**2** or AC**3** according to practical applications or design requirements.

(28) Please refer to FIG. 5. FIG. 5 is a circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit **1**, **2** or **5** of the embodiment of the present invention. The attenuator circuit **403** of FIG. 5 may correspond to the attenuator circuit **40** of FIG. 1 or FIG. 3, or the attenuator circuit **50** of FIG. 2 or FIG. 3.

(29) The attenuator circuit **403** includes an input terminal IN for receiving the input signal or the amplified input signal, an output terminal OUT, a reference voltage terminal REF, resistor-switch circuits RS**0**, RS**1**, and RS**2**. The resistor-switch circuit RS**0** includes a first terminal coupled to one of the input terminal IN and the output terminal OUT, a second terminal coupled to the other one of the input terminal IN and the output terminal OUT, switches SW**60** and SW**631**, and resistors R**631** and RA**631**. The resistor-switch circuit RS**1** includes a first terminal coupled to one of the input terminal IN and the reference voltage terminal REF, a second terminal coupled to the other one of the input terminal IN and the reference voltage terminal REF, resistors R**611** and RA**611**, and a switch SW**611**. The resistor-switch circuit RS**2** includes a first terminal coupled to one of the output terminal OUT and the reference voltage terminal REF, a second terminal coupled to the other one of the output terminal OUT and the reference voltage terminal REF, resistors R**621** and

RA621 and a switch SW621.

(30) The first terminal of the switch SW60 is coupled to the input terminal IN of the attenuator circuit 403. The second terminal of the switch SW60 is coupled to the output terminal OUT of the attenuator circuit 403. The control terminal of the switch SW60 is for receiving the control signal CS60. The control signal CS60 can turn on or turnoff the switch SW60. The first terminal of the resistor R611 is coupled to the input terminal IN of the attenuator circuit 403. The second terminal of the resistor R611 is coupled to the first terminal of the switch SW611. The second terminal of the switch SW611 is coupled to the reference voltage terminal REF. The control terminal of the switch SW611 is for receiving the control signal CS61. The first terminal of the resistor RA611 is coupled to the second terminal of the resistor R611 and the first terminal of the switch SW611. The second terminal of the resistor RA611 is coupled to the second terminal of the switch SW611. The first terminal of the resistor R621 is coupled to the output terminal OUT of the attenuator circuit 403. The second terminal of the resistor R621 is coupled to the first terminal of the switch SW621. The second terminal of the switch SW621 is coupled to the reference voltage terminal REF, and the control terminal of the switch SW621 is for receiving the control signal CS61. The first terminal of the resistor RA621 is coupled to the second terminal of the resistor R621 and the first terminal of the switch SW621. The second terminal of the switch SW621 is coupled to the second terminal of the switch SW621. The first terminal of the resistor R631 is coupled to the input terminal IN of the attenuator circuit 403, and the second terminal of the resistor R631 is coupled to the first terminal of the switch SW631. The second terminal of the switch SW631 is coupled to the output terminal OUT of the attenuator circuit 403. The control terminal of the switch SW631 is for receiving the control signal CS61. The first terminal of the resistor RA631 is coupled to the second terminal of the resistor R631 and the first terminal of the switch SW631. The second terminal of the resistor RA631 is coupled to the second terminal of the switch SW631. The control signal CS61 can turn on and turn off the switches SW611-SW631. The control signals CS60 and CS61 can be provided by an internal circuit in the attenuator circuit 403 or an external circuit outside of the attenuator circuit 403.

(31) For example, by turning on the switch SW60 and turning off the switches SW611-SW631, the attenuator circuit 403 can provide the first attenuation. Alternatively, turning off the switch SW60 and turning on the switches SW611-SW631 can form the first π -type attenuator. The first π -type attenuator equivalently includes resistors R611-R631 and switches SW611-SW631, and the attenuator circuit 403 can provide the second attenuation. Alternatively, the switches SW60, SW611-SW631 can be turned off to form the second π -type attenuator. The second π -type attenuator equivalently includes resistors R611-R631 and RA611-RA631, and the attenuator circuit 403 can provide the third attenuation. The third attenuation may be greater than the second attenuation, and the second attenuation may be greater than the first attenuation. That is, resistors R611, R621 and R631 are shared in the first and second π -type attenuators, so occupied areas of the attenuator circuit are reduced.

(32) Further, the on-resistance of the switch SW60 can be chosen based on the first attenuation. The on-resistance of the switches SW611-SW631 and/or the resistance of the resistors R611-R631 can be chosen based on the second attenuation. The resistance of the resistors RA611-RA631 are chosen based on the third attenuation and the resistance of the resistors R611 to R631. That is, the switches SW611-SW631 can operate to switch the attenuator circuit 403 between the first π -type attenuator and the second π -type attenuator to provide corresponding attenuations. In other embodiments, the overall equivalent resistance of the switch SW611 and the resistor R611 and the overall equivalent resistance of the switch SW621 and the resistor R621 can be substantially the same. In another example, the first attenuation, the second attenuation and the third attenuation can be 0 dB, 5 dB and 6 dB respectively, and the resistance of the switch SW60 can be less than the overall equivalent resistance of the switch SW631 and the resistor R631; the overall equivalent resistance of switch SW631 and the resistor R631 can be less than the overall equivalent resistance

of the resistors **R631** and **RA631**. In this embodiment, the switches **SW60**, **SW611-SW631** can have appropriate on-resistance by carefully choosing the channel width-to-length ratio of the switches **SW60** and **SW611-SW631**.

(33) Please refer back to FIG. 4. When the control signals **CS40-CS4n** turn off the switches **SW40-SW43n** respectively, the attenuator circuit **401** is disabled. At this time, the output terminal of the amplifier **201** and the output terminal **12** of the amplifier circuit **1** are electrically disconnected. In other words, the output terminal **12** of the amplifier circuit **1** would not output the output signal **Sout** because the amplifier circuit **1** is now a zero gain amplifier. However, in the case where the attenuator circuit **401** being replaced by the attenuator circuit **403**, when the control signals **CS60-CS61** turn off the switches **SW60-SW631**, the attenuator circuit **403** can form a π -type attenuator. That is, the attenuator circuit **403** is still enabled, and the output terminal of the amplifier **201** and the output terminal **12** of the amplifier circuit **1** can be electrically connected through the attenuator circuit **403**. In other words, the output signal **Sout** can be generated from the attenuator signal **Sat1**, and the output terminal **12** of the amplifier circuit **1** can still output the output signal **Sout**. In this way, the gain of the amplifier circuit **1** is associated with the attenuation of the amplified input signal **Sin'**.

(34) It should be noted that since the attenuator circuit **403** has a plurality of different π -type attenuators (e.g., the first π -type attenuator and the second π -type attenuator), it can be regarded as having a substantially symmetrical circuit structures. In each of the circuit structure, the difference between the impedance at the input terminal **IN** and the impedance at the output terminal **OUT** of the attenuator circuit **403** can be within a predetermined range. The predetermined range may be $\pm 10\%$. In other embodiments, the impedance at the input terminal **IN** of the attenuator circuit **403** and the impedance at the output terminal **OUT** of the attenuator circuit **403** may be substantially equal. Substantially equal means conjugate matching between the impedance at the input terminal **IN** and the impedance at the output terminal **OUT** of the attenuator circuit **403**. That is, the impedance at the input terminal **IN** and the impedance at the output terminal **OUT** of the attenuator circuit **403** are complex conjugate of each other.

(35) Further, the impedance at the input terminal **IN** of the attenuator circuit **403** can be the equivalent impedance looking from the input terminal **IN** into the internal of the attenuator circuit **403**. The impedance at the output terminal **OUT** of the attenuator circuit **403** can be the equivalent impedance looking from the internal towards the output terminal **OUT** of the attenuator circuit **403**. In addition, the difference between the equivalent impedance looking from the matching circuit **60** into the attenuator circuit **403** and the equivalent impedance looking from the attenuator circuit **403** towards the output terminal **12** of the amplifier circuit **1** can also be within a predetermined range.

(36) The relationship between the various attenuations and the switches **SW60** and **SW611-SW631** described in the above paragraphs is summarized in Table 1.

(37) TABLE-US-00001 TABLE 1 SW60 SW611-SW631 First attenuation ON OFF Second attenuation OFF ON Third attenuation OFF OFF Third attenuation > Second attenuation > First attenuation

(38) Please refer to FIG. 6. FIG. 6 is another circuit diagram of the attenuator circuit in the gain-adjustable amplifier circuit **1**, **2** or **3** of the embodiment of the present invention. The attenuator circuit **404** of FIG. 6 may correspond to the attenuator circuit **40** of FIG. 1 or FIG. 3, or the attenuator circuit **50** of FIG. 2 or FIG. 3.

(39) The major difference between the attenuator circuit **404** of FIG. 6 and the attenuator circuit **403** is that the resistor-switch circuits **RS1**, **RS2**, and **RS0** of the attenuator circuit **404** further include at least resistors **R61n**, **R62n** and **R63n** respectively, and switches **SW61n**, **SW62n** and **SW63n** respectively. In the above symbols, "n" may be a positive integer greater than 1. As shown in FIG. 6, the first terminal of the resistor **R61n** is coupled to the second terminal of the resistor **R61** (n-1) and the first terminal of the switch **SW61** (n-1). The second terminal of the resistor **R61n** is coupled to the first terminal of the switch **SW61n** and the first terminal of the resistor

RA611. The second terminal of the switch **SW61n** is coupled to the reference voltage terminal **REF**. The control terminal of the switch **SW61n** is for receiving the control signal **CS6n**. The first terminal of the resistor **R62n** is coupled to the second terminal of the resistor **R62** ($n-1$) and the first terminal of the switch **SW62** ($n-1$). The second terminal of the resistor **R62n** is coupled to the first terminal of the switch **SW62n** and the first terminal of the resistor **RA621**. The second terminal of the switch **SW62n** is coupled to the reference voltage terminal **REF**. The control terminal of the switch **SW62n** is for receiving the control signal **CS6n**. The first terminal of the resistor **R63n** is coupled to the second terminal of the resistor **R63** ($n-1$) and the first terminal of the switch **SW63** ($n-1$). The second terminal of the resistor **R63n** is coupled to the first terminal of the switch **SW63n** and the first terminal of the resistor **RA631**. The second terminal of the switch **SW63n** is coupled to the output terminal **OUT** of the attenuator circuit **404**. The control terminal of the switch **SW63n** is for receiving the control signal **CS6n**. The control signal **CS6n** can turn on or turn off the switches **SW61n-SW63n**. The control signals **CS60**, **CS61** to **CS6n** may be provided by an internal circuit of the attenuator circuit **404** or by an external circuit outside of the attenuator circuit **404**.

(40) Taking $n=2$ as an example, when turning on the switch **SW60** and turning off the switches **SW611-SW63n**, the attenuator circuit **404** can provide the first attenuation. Alternatively, when turning off the switches **SW60**, **SW61n-SW63n**, and turning on the switches **SW611-SW631**, the attenuator circuit **404** can form a first π -type attenuator. The first π -type attenuator includes the resistors **R611-R631**, the switches **SW611-SW631**. In this configuration, the attenuator circuit **404** can provide second attenuation. Alternatively, when turning off the switches **SW60**, **SW611-SW631**, and turning on the switches **SW61n-SW63n**, the attenuator circuit **404** can form a second π -type attenuator. The second π -type attenuator includes the resistors **R61n-R63n**, the switches **SW61n-SW63n**. In this configuration, the attenuator circuit **404** can provide third attenuation. Alternatively, turning off the switches **SW60**, **SW611-SW63n** can form a third π -type attenuator. The third π -type attenuator includes resistors **RA611-RA631**. In this configuration, the attenuator circuit **404** can provide the fourth attenuation. The fourth attenuation may be greater than the third attenuation; the third attenuation may be greater than the second attenuation, and the second attenuation may be greater than the first attenuation.

(41) Further, the on-resistance of the switch **SW60** can be chosen based on the first attenuation. The on-resistance of the switches **SW611-SW631** and/or the resistance of the resistors **R611-R631** can be chosen based on the second attenuation. The on-resistance of the switches **SW61n-SW63n** and/or the resistance of the resistors **R61n-R63n** can be chosen based on the third attenuation and the resistance of the resistors **R611-R631**. The resistance of the resistors **RA611-RA631** can be chosen based on the fourth attenuation and the resistance of the resistors **R611-R63n**, and so on. That is, the switches **SW611-SW63n** not only can switch between the first π -type attenuator, the second π -type attenuator and the third π -type attenuator, but also can provide the corresponding attenuation. In other embodiments, the overall equivalent resistance of the switch **SW611** and the resistor **R611** can be the same as the overall equivalent resistance of the switch **SW621** and the resistor **R621**. For example, the first attenuation, the second attenuation, the third attenuation and the fourth attenuation can be 0 dB, 5 dB, 10 dB and 11 dB respectively. The overall equivalent resistance of the switch **SW60** can be less than the overall equivalent resistance of the switch **SW631** and the resistor **R631**, the overall equivalent resistance of the switch **SW631** and the resistor **R631** can be less than the overall equivalent resistance of the switch **SW63n** and the resistors **R63n** and **R631**, and the overall equivalent resistance of the switch **SW63n** and the resistors **R63n** and **R631** can be less than the overall equivalent resistance of the resistors **R631**, **R63n** and **RA631**, and so on. In this embodiment, the channel width-to-length ratios of the switches **SW60**, **SW611-SW63n** can be chosen so that the switches **SW60**, **SW611-SW63n** have appropriate on-resistance.

(42) It should be noted that since the attenuator circuit **404** can form a plurality of different π -type

attenuators, which can be regarded as having substantially symmetrical circuit structures. In each of the circuit structures, the difference between the impedance at the input terminal IN of the attenuator circuit **404** and the impedance at the output terminal OUT of the attenuator circuit **404** is within a predetermined range. The predetermined range may be $\pm 10\%$. In other embodiments, the impedance at the input terminal IN of the attenuator circuit **404** and the impedance at the output terminal OUT of the attenuator circuit **404** may be substantially equal. Substantially equal means conjugate matching between the impedance at the input terminal IN of the attenuator circuit **404** and the impedance at the output terminal OUT of the attenuator circuit **404**. That is, the impedance at the output terminal OUT of the attenuator circuit **404** and the impedance at the input terminal IN of the attenuator circuit **404** are complex conjugate of each other.

(43) Further, the impedance at the input terminal IN of the attenuator circuit **404** can be the equivalent impedance looking from the input terminal IN into the internal of the attenuator circuit **404**. The impedance at the output terminal OUT of the attenuator circuit **404** can be the equivalent impedance looking from the internal towards the output terminal OUT of the attenuator circuit **404**. In addition, the difference between the equivalent impedance looking from the matching circuit **60** into the attenuator circuit **404** and the equivalent impedance looking from the attenuator circuit **404** towards the output terminal **12** of the amplifier circuit **1** can also be within a predetermined range.

(44) When $n=2$, the relationship between the various attenuations and the switches SW**60**, SW**611-SW631** and SW**61n-SW63n** described in the above paragraphs is summarized in Table 2.

(45) TABLE-US-00002

TABLE 2	SW60	SW611-SW631	SW61n-SW63n	First attenuation
ON	OFF	OFF	OFF	Second attenuation
OFF	ON	OFF	OFF	Third attenuation
OFF	OFF	ON	OFF	Fourth attenuation
OFF	OFF	OFF	ON	Fourth attenuation

OFF OFF OFF Fourth attenuation > Third attenuation > Second attenuation > First attenuation

(46) Please note that the integer “n” can be greater than 2. As such, there can be as many chain coupled switches SW**61n-SW63n** and resistors R**61n-R63n** in the attenuator circuit **404** as possible. This configuration allows the attenuator circuit **404** to provide more choice of attenuations for flexible implementation.

(47) Please refer to FIG. 7. FIG. 7 is another circuit diagram of the attenuator circuit **40** of the gain-adjustable amplifier circuit **1**, **2** or **3** of the embodiment of the present invention. The attenuator circuit **405** of FIG. 7 may correspond to the attenuator circuit **40** of FIG. 1 or FIG. 3 or attenuator circuit **50** of FIG. 2 or FIG. 3.

(48) The major difference between the attenuator circuit **404** and the attenuator circuit **405** is that the attenuator circuit **405** may further include a switch SWP. The switch SWP may be coupled to between the second terminal of the resistor-switch circuits RS**1** and/or RS**2**, and the reference voltage terminal REF. The control terminal can receive a control signal CSp. The control signal CSp can turn on or turn off the switch SWP.

(49) Taking $n=2$ as an example, when turning on the switch SW**60** and turning off the switches SW**611-SW63n** and the switch SWP, the attenuator circuit **405** can provide the first attenuation. Alternatively, when turning off the switches SW**60**, SW**61n-SW63n**, and turning on the switches SW**611-SW631** and the switch SWP, the attenuator circuit **405** can form a first π -type attenuator. The first π -type attenuator includes the resistors R**611-R631**, the switches SW**611-SW631** and the switch SWP. In this configuration, the attenuator circuit **405** can provide second attenuation. Alternatively, when turning off the switches SW**60**, SW**611-SW631**, and turning on the switches SW**61n-SW63n** and the switch SWP, the attenuator circuit **405** can form a second π -type attenuator. The second π -type attenuator includes the resistors R**611-R63n**, the switches SW**61n-SW63n** and the switch SWP. In this configuration, the attenuator circuit **405** can provide third attenuation. Alternatively, when turning off the switches SW**60** and SW**611-SW63n** and turning on the switch SWP, the attenuator circuit **405** can form a third π -type attenuator. The third π -type attenuator includes the resistors R**611-R631** and RA**611-RA631** and the switch SWP. In this configuration, the attenuator circuit **405** can provide the fourth attenuation. The fourth attenuation may be greater than the third attenuation; the third attenuation may be greater than the second attenuation, and the

second attenuation may be greater than the first attenuation.

(50) When the attenuator circuit **405** provides the first attenuation, the switches SW**611**-SW**61n**, SW**621**-SW**62n**, and SWP are equivalent to capacitors with their respective off-capacitance. The equivalent circuit can be regarded as the capacitors with the off-capacitance of the switches SW**611**-SW**61n** and the switches SW**621**-SW**61n** coupled connected in series with the capacitor with the off-capacitance of the switch SWP. Therefore, the switch SWP can reduce the overall capacitance of the attenuator circuit **405**, thereby avoiding load effect. When the attenuator circuit **405** provides the second attenuation, the third attenuation or the fourth attenuation, the turned-on switch SWP is equivalent to a resistor. The switch SWP can provide flexibility to the resistance of the first π -type attenuator, the second π -type attenuator or the third π -type attenuator. Further, the on-resistance of the switches SW**611**-SW**631** and SWP and/or the resistance of the resistors R**611**-R**631** can be chosen based on the second attenuation. The resistance of the resistors R**611**-R**631** can be chosen based on the third attenuation. The on-resistance of the switch SWP can be chosen based on the on-resistance of the switches SW**61n**-SW**63n** and/or the resistance of the resistors R**61n**-R**63n**. The resistance of the resistors R**611**-R**63n**, the on-resistance of the switch SWP and the resistance of the resistors RA**611**-RA**631** can be chosen based on the fourth attenuation and so on.

(51) When $n=2$, the relationship between the various attenuations and the switches SW**60**, SW**611**-SW**631**, SW**61n**-SW**63n**, and SWP described in the above paragraphs is summarized in Table 3.

(52) TABLE-US-00003

	TABLE 3	SW60	SW611-SW631	SW61n-SW63n	SWP	First attenuation							
ON	OFF	OFF	OFF	Second attenuation	OFF	ON	OFF	ON	Third attenuation	OFF	OFF	ON	ON
Fourth attenuation	OFF	OFF	OFF	ON	Fourth attenuation	>	Third attenuation	>	Second attenuation	>	First attenuation		

(53) Please note that the integer “n” can be greater than 2. As such, there can be as many chain coupled switches SW**61n**-SW**63n** and resistors R**61n**-R**63n** in the attenuator circuit **405** as possible. This configuration allows the attenuator circuit **405** to provide more choice of attenuations for flexible implementation.

(54) Please refer to FIG. 8. FIG. 8 is another circuit diagram of the attenuator circuit **40** of the gain-adjustable amplifier circuit **1**, **2** or **3** of the embodiment of the present invention. The attenuator circuit **406** of FIG. 8 may correspond to the attenuator circuit **40** of FIG. 1 or FIG. 3 or the attenuator circuit **50** of FIG. 2 or FIG. 3.

(55) The major difference between the attenuator circuit **406** and the attenuator circuit **405** is that the attenuator circuit **406** may further include a resistor RP**1**. The resistor RP**1** may be coupled between the switch SWP and the reference voltage terminal REF. The resistor RP**1** can provide more flexibility to choosing the resistance of each π -type attenuator (e.g., the first to third π -type attenuators).

(56) Please refer to FIG. 9. FIG. 9 is another circuit diagram of the attenuator circuit **40** of the gain-adjustable amplifier circuit **1**, **2** or **3** of the embodiment of the present invention. The attenuator circuit **407** of FIG. 9 may correspond to the attenuator circuit **40** of FIG. 1 or FIG. 3, or the attenuator circuit **50** of FIG. 2 or FIG. 3.

(57) The major difference between the attenuator circuit **407** and the attenuator circuit **405** is that the attenuator circuit **407** may further include a resistor RP**2** coupled in parallel to the switch SWP. The resistor RP**2** can provide more flexibility to choosing the resistance of each π -type attenuator (e.g., the first to third π -type attenuators).

(58) In addition, the attenuator circuit **50** shown in FIG. 2 or 3 may be implemented by any one of the attenuator circuits in FIGS. 4 to 9. The implementation detail is not repeated here for brevity.

(59) In the above embodiments, the switches and the transistors mentioned may be field effect transistors (FET) or other types of transistors. When the switches and the transistors are FETs, the first terminal may be either the drain or the source; the second terminal may be the other one of the drain and the source; the control terminal may be the gate.

(60) The gain-adjustable amplifier circuit provided by the various embodiments of the present

invention can expand the gain range of the amplifier circuit and adjust the gain of the amplifier circuit through the attenuator circuits. Moreover, since the attenuator circuit can have substantially symmetrical circuit structure, the attenuator circuit would not easily affect the output impedance matching of the amplifier in the amplifier circuit. In addition, by sharing at least one resistor in different π -type attenuators, occupied areas of the attenuator circuit are reduced.

(61) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An amplifier circuit comprising: an amplifier configured to generate an amplified input signal according to an input signal; and an attenuator circuit coupled to the amplifier, comprising: an input terminal for receiving the input signal or the amplified input signal; an output terminal; a reference voltage terminal; a zeroth resistor-switch circuit comprising: a first terminal; a second terminal, wherein the first terminal is coupled to the input terminal, and the second terminal is coupled to the output terminal; or the first terminal is coupled to the output terminal and the second terminal is coupled to the input terminal; a zeroth switch coupled between the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit; a zeroth resistor coupled between the first terminal of the zeroth resistor-switch circuit and the second terminal of the zeroth resistor-switch circuit; a first resistor coupled between the zeroth resistor and the second terminal of the zeroth resistor-switch circuit; and a first switch comprising: a first terminal coupled to the zeroth resistor and the first resistor; a second terminal coupled to the second terminal of the zeroth resistor-switch circuit; and a control terminal; a first resistor-switch circuit comprising: a first terminal; a second terminal, wherein the first terminal is coupled to the input terminal, and the second terminal is coupled to the reference voltage terminal; or the first terminal is coupled to the reference voltage terminal, and the second terminal is coupled to the input terminal; a zeroth resistor coupled between the first terminal of the first resistor-switch circuit and the second terminal of the first resistor-switch circuit; a first resistor coupled between the zeroth resistor and the second terminal of the first resistor-switch circuit; and a first switch comprising: a first terminal coupled to the zeroth resistor and the first resistor; a second terminal coupled to the second terminal of the first resistor-switch circuit; and a control terminal; and a second resistor-switch circuit comprising: a first terminal; a second terminal, wherein the first terminal is coupled to the output terminal, and the second terminal is coupled to the reference voltage terminal; or the first terminal is coupled to the reference voltage terminal, and the second terminal is coupled to the output terminal; a zeroth resistor coupled between the first terminal of the second resistor-switch circuit and the second terminal of the second resistor-switch circuit; a first resistor coupled between the zeroth resistor and the second terminal of the second resistor-switch circuit; and a first switch comprising: a first terminal coupled to the zeroth resistor and the first resistor; a second terminal coupled to the second terminal of the second resistor-switch circuit; and a control terminal.
2. The amplifier circuit of claim 1, wherein in a first mode, the attenuator circuit has a first attenuation and is configured to turn on the zeroth switch of the zeroth resistor-switch circuit and turn off the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit.
3. The amplifier circuit of claim 2, wherein in a second mode, the attenuator circuit has a second attenuation and is configured to turn off the zeroth switch of the zeroth resistor-switch circuit and turn on the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit.
4. The amplifier circuit of claim 3, wherein in a third mode, the attenuator circuit has a third

attenuation and is configured to turn off the zeroth switch of the zeroth resistor-switch circuit and turn off the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit.

5. The amplifier circuit of claim 4, wherein the third attenuation is greater than the second attenuation, and the second attenuation is greater than the first attenuation.

6. The amplifier circuit of claim 1, wherein a difference between an impedance at the input terminal of the attenuator circuit and an impedance at the output terminal of the attenuator circuit is within $\pm 10\%$.

7. The amplifier circuit of claim 1, wherein: the zeroth resistor-switch circuit further comprising: a second resistor coupled between the first resistor and the second terminal of the zeroth resistor-switch circuit; and a second switch comprising: a first terminal coupled to the first resistor and the second resistor; a second terminal coupled to the second terminal of the zeroth resistor-switch circuit; and a control terminal; the first resistor-switch circuit further comprising: a second resistor coupled between the first resistor and the second terminal of the first resistor-switch circuit; and a second switch comprising: a first terminal coupled to the first resistor and the second resistor; a second terminal coupled to the second terminal of the first resistor-switch circuit; and a control terminal; and the second resistor-switch circuit further comprising: a second resistor coupled between the first resistor and the second terminal of the second resistor-switch circuit; and a second switch comprising: a first terminal coupled to the first resistor and the second resistor; a second terminal coupled to the second terminal of the second resistor-switch circuit; and a control terminal.

8. The amplifier circuit of claim 7, wherein in a first mode, the attenuator circuit has a first attenuation and is configured to turn on the zeroth switch of the zeroth resistor-switch circuit, and turn off the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit, and turn off the second switch of the zeroth resistor-switch circuit, the second switch of the first resistor-switch circuit and the second switch of the second resistor-switch circuit.

9. The amplifier circuit of claim 8, wherein in a second mode, the attenuator circuit has a second attenuation and is configured to turn off the zeroth switch of the zeroth resistor-switch circuit, and turn on the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit, and turn off the second switch of the zeroth resistor-switch circuit, the second switch of the first resistor-switch circuit and the second switch of the second resistor-switch circuit.

10. The amplifier circuit of claim 9, wherein in a third mode, the attenuator circuit has a third attenuation and is configured to turn off the zeroth switch of the zeroth resistor-switch circuit, and turn off the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit, and turn on the second switch of the zeroth resistor-switch circuit, the second switch of the first resistor-switch circuit and the second switch of the second resistor-switch circuit.

11. The amplifier circuit of claim 10, wherein in a fourth mode, the attenuator circuit has a fourth attenuation and is configured to turn off the zeroth switch of the zeroth resistor-switch circuit, and turn off the first switch of the zeroth resistor-switch circuit, the first switch of the first resistor-switch circuit and the first switch of the second resistor-switch circuit, and turn off the second switch of the zeroth resistor-switch circuit, the second switch of the first resistor-switch circuit and the second switch of the second resistor-switch circuit.

12. The amplifier circuit of claim 11, wherein: the fourth attenuation is greater than the third attenuation; the third attenuation is greater than the second attenuation; and the second attenuation is greater than the first attenuation.

13. The amplifier circuit of claim 7 further comprising a reference switch coupled between the first resistor-switch circuit, the second resistor-switch circuit and the reference voltage terminal.

14. The amplifier circuit of claim 13 further comprising a reference resistor coupled in series with the reference switch.
 15. The amplifier circuit of claim 13 further comprising a reference resistor coupled in parallel with the reference switch.
 16. The amplifier circuit of claim 7, wherein a difference between an impedance at the input terminal of the attenuator circuit and an impedance at the output terminal of the attenuator circuit is within $\pm 10\%$.
 17. The amplifier circuit of claim 1, wherein: the first terminal of the zeroth resistor-switch circuit is coupled the input terminal; the second terminal of the zeroth resistor-switch circuit is coupled to the output terminal; the first terminal of the first resistor-switch circuit is coupled to the input terminal; a second terminal of the first resistor-switch circuit is coupled to the reference voltage terminal; the first terminal of the second resistor-switch circuit is coupled to the input terminal; and a second terminal of the second resistor-switch circuit is coupled to the reference voltage terminal.
 18. The amplifier circuit of claim 1 further comprising a reference switch coupled between the first resistor-switch circuit, the second resistor-switch circuit and the reference voltage terminal.
 19. The amplifier circuit of claim 18 further comprising a reference resistor coupled in series with the reference switch.
 20. The amplifier circuit of claim 18 further comprising a reference resistor coupled in parallel with the reference switch.
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